the surrounding environment. Organisms which are unable to face the challenges, are unfit to survive and disappear. The process of selection of organisms with favourable variation is called as natural selection.

v. **Origin of species**
According to Darwin, **new species originates by the gradual accumulation of favourable variations** for a number of generations.

19.4 Variation

Sexual reproduction, which involves meiosis helps in **recombination of genes** during gametic fusion. This leads to differences in the phenotype of the offspring from its parents. These differences are called variation.

Variation is the difference found among individuals of the same species and the offspring of the same parent. **Variation** is the **raw material** which plays an important **role in evolution**. Evolution would not be possible without variation.

Types of variations

Somatic variation: These are the variations which affect the body (somatic) cells of the organisms. They are **not heritable**. They occur due to environmental factors.

Germinal variation: These variations are produced in germ cells of an organism. They are **inherited**. They may be present in ancestors or may occur suddenly. They are classified into two types:

1. Continuous variation
2. Discontinuous variation

Continuous variation: These are small variations which occur among individuals of a population. They are also called as **fluctuating variations**. They occur by **gradual accumulation** in a population. e.g. skin colour, height and weight of an individual, colour of eye, etc.

Discontinuous variation: These are **sudden changes** which occur in an organism due to mutations. They do not have any intermediate forms. These large variations are not useful for evolution. e.g. short legged Ancon sheep, six or more digits (fingers) in human, etc.

Discontinuous variation form the basis for **Mutation theory** proposed by **De Vries**.

Relationship between Mutation and Variation

Mutation and Variation are two events involved in the process of evolution. Mutation arises due to errors occurring in DNA during replication or exposure to UV rays or chemicals. Mutation leads to variation. It brings about changes in a single individual.

19.5 Paleobotany

Paleobotany is derived from Greek words *paleon* that means "old" and *botany* the study of plants. It is the branch of paleontology that deals with **recovery and identification of plant remains of geological past**.

A plant fossil is any preserved part of a plant that has died long back. Fossils may be a prehistoric impression that may be hundred to millions of years old. Majority of the plant fossils are disarticulated parts of plants, it is rare to find plants to be preserved as whole.

Importance of fossils

i They throw light on phylogeny and evolution of plants.
ii Fossil plants give a historical approach to plant kingdom.
iii Fossils are useful in classification of plants.
iv. Fossil plants can be used in the field of descriptive and comparative anatomy.

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